Hyperloop Improvements?

By: Jonathan Distler

Cornell Hyperloop

H-Numbering System

H-Number	Name:	Subsystem:	Designer:	Approver:
H-0001	Chassis	Structures	Jonathan D.	Carter
H-0002				
H-0003				
H-0004				

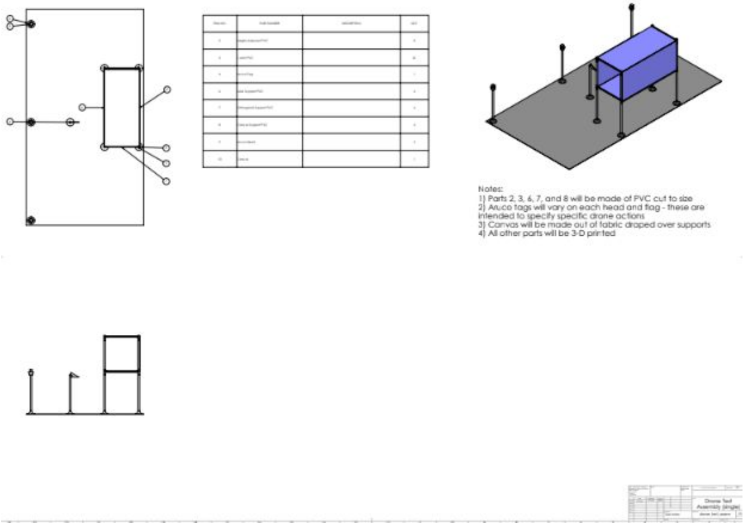
Part/Assembly name

Subsystem - relates to has to approve it

Designer and Approver - in this example it could be manufactured

- Each custom part and assembly gets assigned an "H-Number" (McMaster-Carr parts without modifications don't require an "H-Number")
- All files are stored in a general Hyperloop folder (rather than subfolders)
- In order to be manufactured (e.g. 3D printed, bought), the part/assembly needs to be designed in CAD and then approved by a team-lead

Drawing/BoM for Each Assembly



- Each assembly should have an associated drawing (simple drag-and-drop with at a minimum one projected view)
    - Best practice for industry
- Each drawing should have an associated BoM (“Insert”->”BoM”)
    - Will make PO’s move faster without redundancy

Constraint Format

Subsystem Name:	Space Constraint:	Cost Constraint:	Physical Demands of System (e.g. applied pressure, required force output):
LIM	Needs to fit in 1' x 1' bounding box	Needs to be <$500	Needs to provide X N of force within Y s of triggering

- CDR/PDR should have a required "Constraint" layout - outlining what the main challenges are (e.g. size, cost, etc.), so that everyone is clear on what they're designing around

- Formatting is very much up to change

Learning Rate

$$y = y_0 e^{kt}$$

- **y** is the *__accumulated knowledge of the Hyperloop system__*, **k** is the *__change in knowledge from iteration__*, and **t** is the *__frequency of iteration__*

- Two variables to "play" with: *t and k*
  - If we **iterate quicker** (with less drastic changes) we will learn more

- Money shouldn't be as much of a consideration as **iteration/"tick rate"**
  - Should prioritize buying

THANK YOU